Name:___________

A2 Chemistry Unit 3

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)

- Calculation errors (forgetting to square numbers, etc.)

- Not reading questions carefully

- Not answering the full question (describe vs. explain)

- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2017	18	24	31	38	45	63
2018	18	25	32	40	48	66
2019	18	25	33	41	49	65
2022	5	9	14	22	30	58
2023	10	17	24	31	39	71

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL – NEW

1410U30-1

S17-1410U30-1

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

TUESDAY, 13 JUNE 2017 – AFTERNOON

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 6.**	**10**	
Section B	**7.**	**7**	
	8.	**10**	
	9.	**15**	
	10.	**11**	
	11.	**10**	
	12.	**17**	
	Total	**80**	

1410U301
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**(*b*).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

JUN171410U30101

Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. When aqueous iodide ions are added to an aqueous solution of lead(II) nitrate, a precipitate is formed.

(a) Give the colour of the precipitate. [1]

....

(b) Write an **ionic** equation for the reaction that occurs, including state symbols. [1]

....

2. Give an example of a transition metal used as a catalyst. You should name the metal and identify the reaction it catalyses. [1]

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....

3. Calcium chloride, $CaCl_2$, is soluble in water. Use the values below to explain why this is the case. [2]

Equation 1 $\quad Ca^{2+}(g) + 2Cl^{-}(g) \longrightarrow CaCl_2(s) \quad \Delta H^{\theta} = -2237\ kJ\ mol^{-1}$

Equation 2 $\quad Ca^{2+}(g) + 2Cl^{-}(g) + aq \longrightarrow CaCl_2(aq) \quad \Delta H^{\theta} = -2378\ kJ\ mol^{-1}$

....

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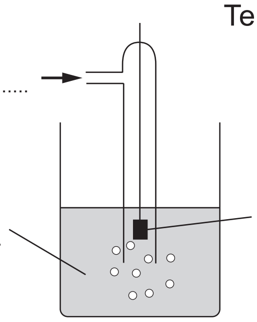
Examiner only

4. Label the diagram of the standard hydrogen electrode below, including all relevant conditions. [2]

Temperature = ...



5. Reaction rates can be measured using an iodine clock reaction. The data below shows the results obtained by a student in an iodine clock experiment. The concentration of iodide ions was the only difference between the experiments.

Concentration of iodide, $[I^-]$ / mol dm^{-3}	Time taken for colour change / s				Mean time taken / s
	1st run	2nd run	3rd run	4th run	
0.200	35	46	36	34	
0.400	18	17	18	17	

(a) Complete the table with appropriate values of mean time taken. [1]

(b) Calculate the rate of reaction for these two concentrations. [1]

Rate for 0.200 mol dm^{-3} $[I^-]$ = ... s^{-1}

Rate for 0.400 mol dm^{-3} $[I^-]$ = ... s^{-1}

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03

Examiner only

6. Place the following substances in order of increasing entropy under standard conditions. [1]

nitrogen gas copper metal water air

Lowest .... *Highest*

10

Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

7. *(a)* Ammonia is an example of a weak base. Describe what is meant by the term *weak base*. [1]

....

....

(b) A mixture of ammonia and ammonium chloride in aqueous solution can be used as a basic buffer solution. Explain what is meant by a *buffer solution* and how this mixture can act as a buffer solution. [3]

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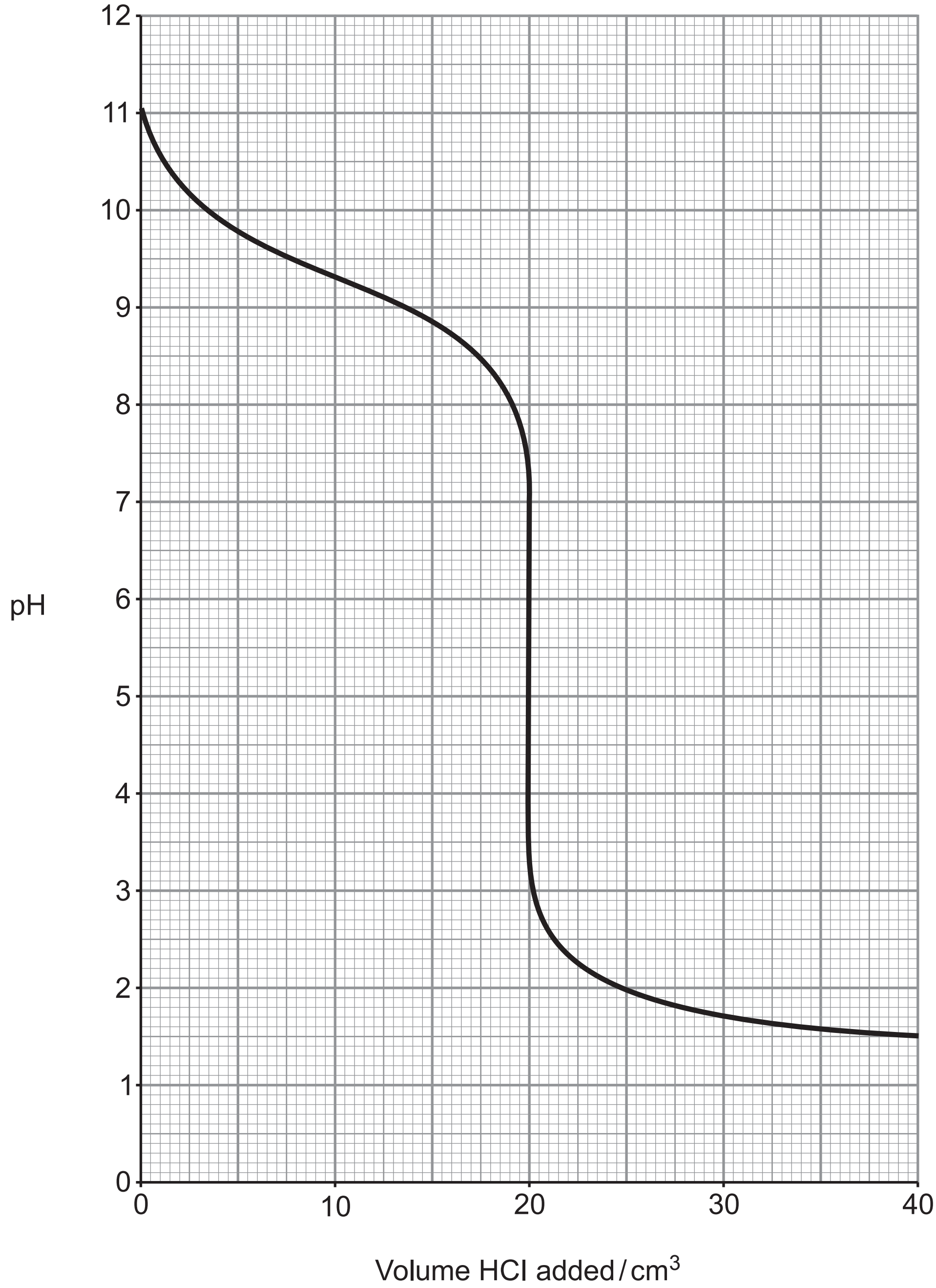
1410U301
05

Examiner only

(c) The pH curve for titration of a solution of ammonia against hydrochloric acid of an equal concentration is given below.



(i) Calculate K_a for the ammonium ion. [2]

K_a = ... mol dm^{-3}

Examiner only

(ii) Select an appropriate indicator for this titration from the list below, giving a reason for your answer. [1]

Indicator	pH range
bromothymol blue	6.0 – 7.6
4-nitrophenol	5.0 – 7.0
methyl yellow	2.9 – 4.0
phenolphthalein	8.2 – 10.0

....

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1410U301 07

7

Examiner only

8. *(a)* Some elements in the *p*-block form compounds where the *p*-block atom does not have eight electrons in its outer shell.

(i) Explain why nitrogen can only form a chloride with eight outer shell electrons, but phosphorus can form a chloride with a different number of outer shell electrons. [2]

You should give the chemical formulae of relevant compounds in your answer.

....

....

....

(ii) Explain why aluminium forms compounds that are electron deficient. Show how one of these compounds can act to gain a full outer shell. [2]

You should give the dot and cross diagrams of the electron deficient species.

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Examiner only

(b) Below is some information about the oxides of two Group 4 elements.

Carbon	There are two common oxides of carbon. Carbon dioxide is an acidic oxide and carbon monoxide can be used as a reducing agent. Both of these are gases with very low boiling temperatures.
Lead	There are two common oxides of lead. Lead(II) oxide is an amphoteric oxide and lead(IV) oxide can be used as an oxidising agent. Both of these are solids that exhibit a large degree of ionic character.

Explain the differences between the oxides of carbon and lead, giving chemical equations to illustrate their acid/base and redox properties. [6 QER]

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Examiner only

9. The reversible reaction shown below was one of the first to be studied in detail in the gas and liquid phases and in solution.

$$N_2O_4 \rightleftharpoons 2NO_2$$

(a) One study of the liquid mixture showed that it contained 0.714 % NO_2 by mass. Calculate how many moles of NO_2 would be present in 25.0 g of this liquid. Give your answer to an appropriate number of significant figures. [2]

$n(NO_2)$ = ... mol

(b) Both N_2O_4 and NO_2 are soluble in a range of solvents and the equilibrium constant, K_c, can be measured in these.

(i) Give the expression for the equilibrium constant, K_c, for this equilibrium, giving its unit if any. [2]

K_c =

Unit = ...

Examiner only

(ii) The equilibrium constants for the N_2O_4/NO_2 equilibrium measured at room temperature in some different solvents are listed below.

Solvent	Equilibrium constant, K_c (Unit not shown)
CS_2	17.8
CCl_4	8.05
$CHCl_3$	5.53
C_2H_5Br	4.79
C_6H_6	2.03
$C_6H_5CH_3$	1.69

I. Samples of 0.4000 mol of N_2O_4 are dissolved separately in 1 dm^3 of each of these solvents at room temperature and the reactions allowed to reach equilibrium. In one solution the concentration of N_2O_4 present at equilibrium is 5.81×10^{-2} mol dm^{-3}. Find the value of the equilibrium constant in this solution and hence identify the solvent. [3]

Solvent ....

II. The Gibbs free energy change, ΔG, of this reaction is different in different solvents. Explain how the data shows this and state, with a reason, which solvent would have the most negative ΔG value for this reaction. [2]

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Examiner only

(c) Nitrogen dioxide is used in the production of nitric acid. It is produced in two stages.

Stage 1 $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$ $\Delta H^\theta = -900\ kJ\ mol^{-1}$

Stage 2 $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $\Delta H^\theta = -115\ kJ\ mol^{-1}$

(i) Explain why the use of a catalyst is essential in industrial processes involving exothermic equilibria such as stage 1. [2]

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(ii) A student attempted to perform stage 2 using a temperature of 450 K and a pressure of 5 atm. He obtained only a small yield. Suggest how the yield could be improved, explaining your reasons. [4]

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15

Examiner only

10. Aqua regia is an acidic liquid that can dissolve unreactive metals such as gold. It can be formed by bubbling hydrogen chloride gas through concentrated nitric acid.

$$HNO_3(l) + 3HCl(g) \longrightarrow NOCl(g) + Cl_2(g) + 2H_2O(l) \qquad \Delta H^\theta = -69 \text{ kJ mol}^{-1}$$

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta$ / kJ mol^{-1}	Standard entropy, S^θ / J K^{-1} mol^{-1}
$HNO_3(l)$	−173	156
$HCl(g)$		187
$NOCl(g)$	53	264
$Cl_2(g)$	0	223
$H_2O(l)$	−286	70

(a) Show that this is a redox reaction. [1]

....

....

(b) State why the enthalpy change of formation of $Cl_2(g)$ is zero. [1]

....

....

(c) Calculate the standard enthalpy change of formation of $HCl(g)$. [2]

$\Delta_f H^\theta$ = ... kJ mol^{-1}

Examiner only

(d) (i) Calculate the free energy change, ΔG^{θ}, for this reaction at 25 °C. [3]

ΔG^{θ} = ... kJ mol^{-1}

(ii) An alternative method of preparing aqua regia is to use concentrated hydrochloric acid with concentrated nitric acid. State, giving a reason, the effect of this on the value of ΔG^{θ} at 25 °C. [1]

....

....

....

(e) (i) The pH of a dilute solution of nitric acid is measured using a pH probe and has a value of 0.3. Calculate the concentration of this solution. [1]

Concentration = ... mol dm^{-3}

(ii) An alternative way of measuring the concentration of the solution is by titration against a standard solution of sodium hydrogencarbonate of concentration 0.500 mol dm^{-3}.

State and explain which of these two methods will give the more precise value for the concentration. [2]

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11

Examiner only

11. A hydrated compound $M_aAl_bZ_c.dH_2O$ contains an *s*-block metal ion, M, and a halide ion, Z.

The following tests were undertaken.

Test	Result
Heating to constant mass	A 0.0200 mol sample lost 1.44 g when heated
Adding concentrated sulfuric acid	Observations included coloured fumes, an orange-brown solution, steamy fumes and a choking gas; there was no smell of rotten eggs
Adding excess silver nitrate to a solution	A 25.0 cm^3 sample of a solution of concentration 0.203 mol dm^{-3} produced a precipitate with a dry mass of 3.814 g
Adding excess sodium hydroxide to a solution	A white precipitate formed which dissolved in excess sodium hydroxide
Elemental analysis	The hydrated compound was found to contain 1.63 % by mass of *s*-block metal and 6.34 % by mass of aluminium

(a) Calculate the value of *d* in the formula $M_aAl_bZ_c.dH_2O$. [2]

d = ...

(b) (i) Identify the halide present in the compound, explaining fully all observations that led to your conclusion. [2]

....

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Examiner only

(ii) Calculate the value of c in the formula $M_aAl_bZ_c.dH_2O$. [2]

c = ...

(c) State what the observations with sodium hydroxide indicate about the acidity/basicity of the aluminium ions. [1]

....

....

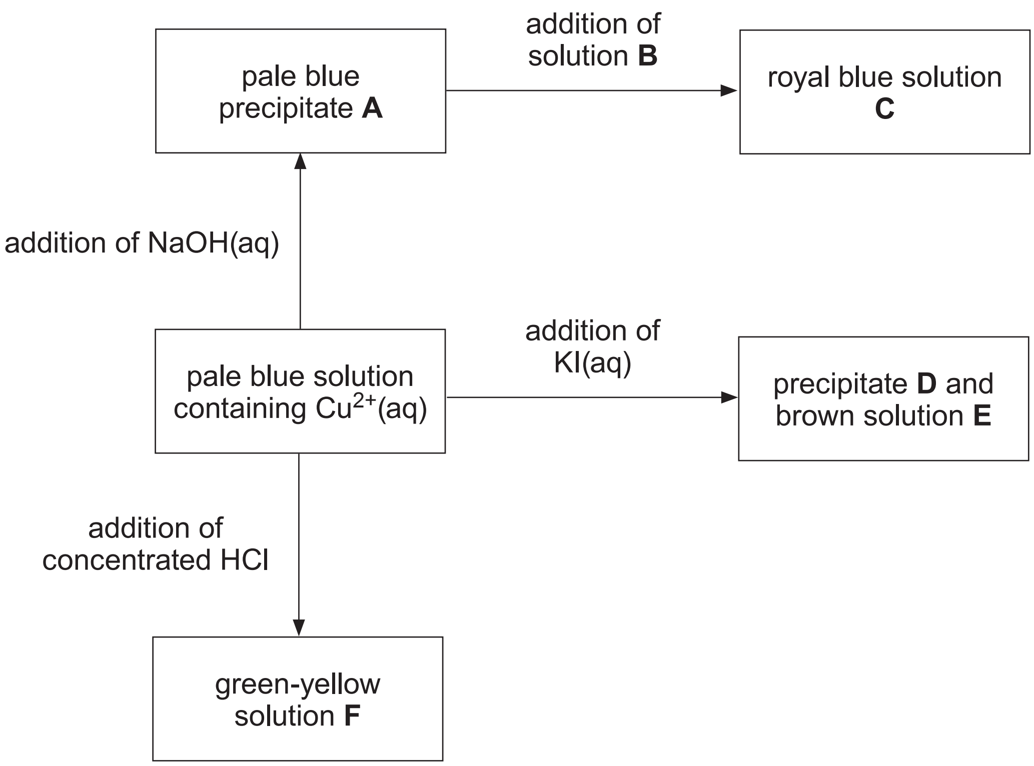
(d) The M_r of the original **hydrated** compound is 425.62. Use all the information above to find the formula of the hydrated compound. Explain your reasoning. [3]

Formula of the **hydrated** compound ....

10

Examiner only

12. *(a)* The diagram below shows some reactions of copper(II) ions in solution.



Identify the species present in each of the following. [3]

A ....

B ....

C ....

D ....

E ....

F ....

Examiner only

(b) Vanadium exhibits the properties of a typical transition element, such as having variable oxidation states and forming coloured complexes. Explain why transition elements have variable oxidation states. [1]

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(c) Blue oxovanadium(IV) cations, VO^{2+}, can be oxidised to yellow oxovanadium(V) ions, VO_2^+, by bromate(V) ions, BrO_3^-.

	Standard electrode potential, E^θ / V
$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$	+1.00
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+1.09
$BrO_3^- + 6H^+ + 6e^- \rightleftharpoons Br^- + 3H_2O$	+1.44

Use these standard electrode potentials to identify the main bromine-containing product formed when excess bromate(V) ions are added to a solution containing VO^{2+} ions. Give reasons for your answer. [3]

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Examiner only

(d) The effect of the concentration of H^+ ions on the rate of the reaction in part *(c)* was studied using the same concentrations and volumes of all other reactants. The following data were collected.

pH	Rate / mol dm^{-3} s^{-1}
1.0	3.287×10^{-3}
1.4	1.308×10^{-3}
1.8	5.193×10^{-4}
2.2	2.071×10^{-4}

Find the order of the reaction with respect to H^+. You **must** show your working. [3]

Order with respect to H^+ = ...

Examiner only

(e) The value of the rate constant, k, for this reaction at two temperatures is given below.

Temperature / K	Rate constant (unit not shown)
288	0.0761
300	0.2873

The activation energy for this reaction is 79.333 kJ mol^{-1} and the Arrhenius equation is:

$$k = Ae^{-\left(\frac{E_a}{RT}\right)}$$

Calculate the value of the rate constant, k, at a temperature of 295 K. [4]

k at 295 K = ...

Examiner only

(f) VO^{2+} ions may also be oxidised to VO_2^+ ions using coloured Ce^{4+}, which is itself reduced to colourless Ce^{3+}. A titration using this reaction does not usually include an indicator and colorimetry is sometimes used.

State why an indicator may not be useful in this reaction and explain how colorimetry can be used to find the end point of the titration. [3]

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17

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL

1410U30-1

S18-1410U30-1

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

TUESDAY, 5 JUNE 2018 – AFTERNOON

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**14**	
	9.	**18**	
	10.	**12**	
	11.	**18**	
	12.	**8**	
	Total	**80**	

1410U301
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

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INFORMATION FOR CANDIDATES

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Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.12***(b)*.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

JUN181410U30101

Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. Give the electronic configuration of a copper atom. [1]

....

2. Excess sodium hydroxide is added to an aqueous solution of chromium(III) chloride, $CrCl_3$. Give the formula of the chromium-containing ion formed. [1]

....

3. *(a)* Write the equation for the reaction of hot concentrated aqueous sodium hydroxide with chlorine. [1]

....

(b) This is an example of a disproportionation reaction. State what is meant by the term *disproportionation*. [1]

....

....

4. Give the observation(s) expected when water is added to $SiCl_4$. [1]

....

....

Examiner only

5. A 25.0 cm^3 sample of a solution containing iodine is titrated against a sodium thiosulfate solution of concentration 0.0200 mol dm^{-3}. This requires 23.25 cm^3 of sodium thiosulfate for complete reaction.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \longrightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

Calculate the concentration of the iodine in the solution in g dm^{-3}. [2]

Concentration = ... g dm^{-3}

6. The enthalpy of solution of sodium chloride is +4 kJ mol^{-1}. Explain why this compound is soluble in water despite this value being positive. [1]

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7. When a solution containing the weak base ammonia is neutralised using the strong acid sulfuric acid, a solution of ammonium sulfate is formed. Suggest a pH for this solution, giving a reason for your answer. [2]

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1410U301
03

10

Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

8. *(a)* Sodium chloride and sodium bromide both react with concentrated sulfuric acid. Describe the observations in both reactions and explain why they are different. [3]

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(b) Fluorine can form the weak acid hydrofluoric acid, HF. This acid has a K_a value of 7.20×10^{-4} mol dm^{-3}.

(i) Write an expression for the K_a of hydrofluoric acid. [1]

(ii) Calculate the pH of a solution of hydrofluoric acid of concentration 0.100 mol dm^{-3}. [3]

pH = ...

Examiner only

(iii) The concentration of a solution of hydrofluoric acid can be found by titrating against sodium hydroxide. Not all acid-base indicators would be suitable for this titration. Explain what features would make an indicator suitable for use in a weak acid-strong base titration. [2]

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(iv) Addition of 12.5 cm^3 of 0.100 mol dm^{-3} sodium hydroxide to 25.0 cm^3 of hydrofluoric acid of concentration 0.100 mol dm^{-3} forms a buffer solution.

I. Explain how this buffer solution works. [3]

....

....

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....

....

....

II. Calculate the pH of this buffer solution. [2]

pH = ...

14

1410U301 05

Examiner only

9. Cobalt forms a range of complex ions. Two of these are $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$.

(a) Draw the structures of the $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$ ions. [2]

(b) Explain why the $[Co(H_2O)_6]^{2+}$ ion is coloured. [3]

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Examiner only

(c) A student was given a pink-coloured solution containing $[Co(H_2O)_6]^{2+}$ ions. Upon addition of hydrochloric acid the solution turned blue as $[CoCl_4]^{2-}$ ions formed according to the equilibrium shown below.

$$[Co(H_2O)_6]^{2+} \quad + \quad 4Cl^- \quad \rightleftharpoons \quad [CoCl_4]^{2-} \quad + \quad 6H_2O$$

Aqueous silver nitrate was added to the solution containing $[CoCl_4]^{2-}$.

State and explain the observation(s) expected. [4]

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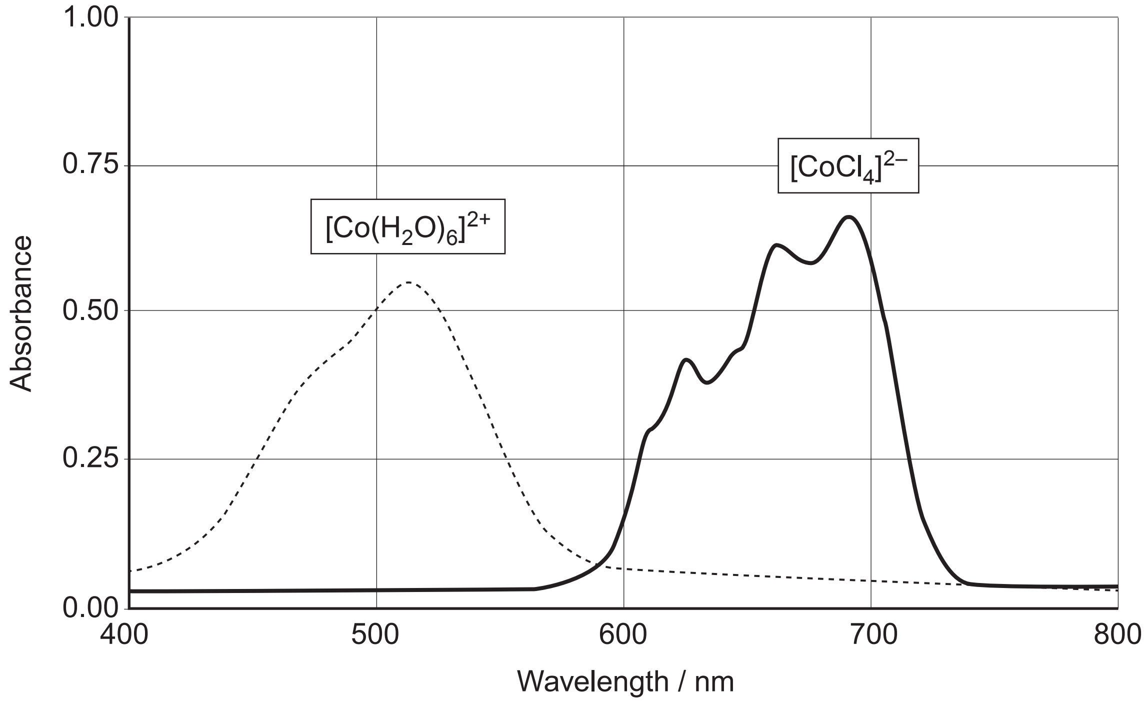
1410U301
07

Examiner only

(d) These two cobalt-containing ions absorb different frequencies of light. The spectra below show the absorbance at different wavelengths of visible light by the two ions.



(i) The maximum absorbance for the $[Co(H_2O)_6]^{2+}$ ion occurs at a wavelength of 518 nm. Calculate the energy of this absorbance in $kJ\,mol^{-1}$. [3]

Energy = ... $kJ\,mol^{-1}$

(ii) A colorimeter was used to study the equilibrium between the two ions.

Suggest, giving a reason, an appropriate wavelength to use for the experiment. [1]

....

....

....

Examiner only

(iii) Write the expression for the equilibrium constant, K_c, for this reaction **giving its unit**. [2]

$$[Co(H_2O)_6]^{2+} \quad + \quad 4Cl^- \quad \rightleftharpoons \quad [CoCl_4]^{2-} \quad + \quad 6H_2O$$

K_c =

Unit ....

(iv) A student carries out an experiment starting with separate non-aqueous solutions of $[Co(H_2O)_6]^{2+}$ and Cl^-.

When the two solutions are mixed the initial concentrations of both $[Co(H_2O)_6]^{2+}$ and Cl^- in the mixture are 0.720 mol dm^{-3}. When the mixture reaches equilibrium the concentration of the water is 0.744 mol dm^{-3}.

Calculate the value of the equilibrium constant, K_c. [3]

K_c = ...

1410U301 09

18

Turn over.

Examiner only

10. *(a)* (i) Tin and carbon are both more stable in oxidation state +4 whilst lead is more stable as +2. Explain the difference in the stabilities of these oxidation states. [1]

....

....

(ii) Lead(IV) oxide can act as an oxidising agent when it reacts with concentrated hydrochloric acid. Write an equation for this reaction. [1]

....

(b) Tin(II) ions can be used to reduce iron(III) ions in **acidic** solution.

$$2Fe^{3+}(aq) + Sn^{2+}(aq) \longrightarrow 2Fe^{2+}(aq) + Sn^{4+}(aq)$$

$[Fe^{3+}]$ / mol dm^{-3}	$[Sn^{2+}]$ / mol dm^{-3}	pH	Initial rate / mol dm^{-3} s^{-1}
0.200	0.100	0	1.4×10^{-2}
0.300	0.200	0	2.8×10^{-2}
0.400	0.100	1	1.4×10^{-3}
0.400	0.200	1	2.8×10^{-3}

(i) Find the rate equation for this reaction. [3]

(ii) Calculate the value of the rate constant, *k*, for this reaction. [2]

k = ...

Examiner only

(iii) A student suggests that the rate determining step for this reaction is:

$$Fe^{3+} + Sn^{2+} \longrightarrow Fe^{2+} + Sn^{3+}$$

State, giving a reason, whether this is a possible rate determining step for this reaction. [2]

....

....

....

(c) The rate of this reaction can be followed using sampling and quenching.

(i) Explain what is meant by *sampling and quenching*. [1]

....

....

....

(ii) A 5.00 cm^3 sample of the solution was analysed by titration against acidified potassium manganate(VII). The sample required 27.20 cm^3 of manganate(VII) solution of concentration 0.00205 $mol\,dm^{-3}$ for complete reaction.

Calculate the concentration of Fe^{2+} in the solution. [2]

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

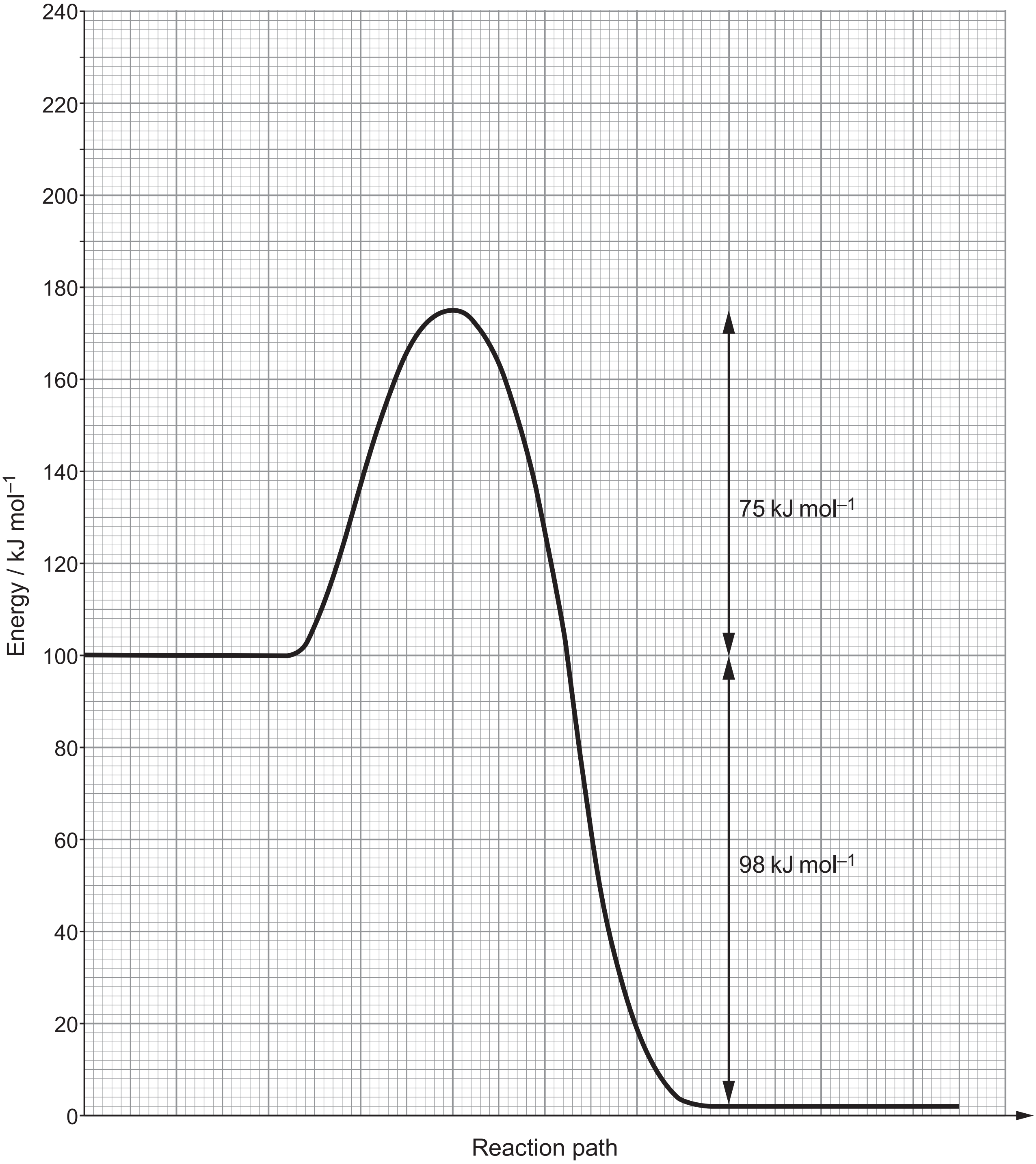
Concentration of Fe^{2+} = ... $mol\,dm^{-3}$

12

11. Hydrogen peroxide, H_2O_2, decomposes slowly at 300 K to form water and oxygen gas.

$$2H_2O_2(l) \longrightarrow 2H_2O(l) + O_2(g)$$

The reaction profile for this reaction is shown below.



Examiner only

(a) The rate of decomposition of hydrogen peroxide can be influenced by a range of catalysts. The activation energy when using MnO_2 as a catalyst is 58 kJ mol^{-1}.

Draw the reaction profile for the catalysed reaction **on the grid opposite**. [1]

(b) Another catalyst was used and this gave a value for the rate constant, *k*, of 1.68 mol^{-1} dm^3 s^{-1} and the frequency factor, *A*, of 1.41×10^4 mol^{-1} dm^3 s^{-1} at a temperature of 300 K.

(i) State the Arrhenius equation. [1]

....

(ii) Calculate the activation energy using this catalyst and hence state whether it is a more effective catalyst than MnO_2. [3]

Activation energy = ... kJ mol^{-1}

....

....

....

(c) The standard enthalpy change of formation, $\Delta_f H^\theta$, of water is –286 kJ mol^{-1}.

Use this information and the graph to calculate the standard enthalpy change of formation of hydrogen peroxide. [2]

$\Delta_f H^\theta(H_2O_2)$ = ... kJ mol^{-1}

(d) State whether you would expect the entropy change for the decomposition of hydrogen peroxide to be positive or negative. Give a reason for your answer. [1]

....

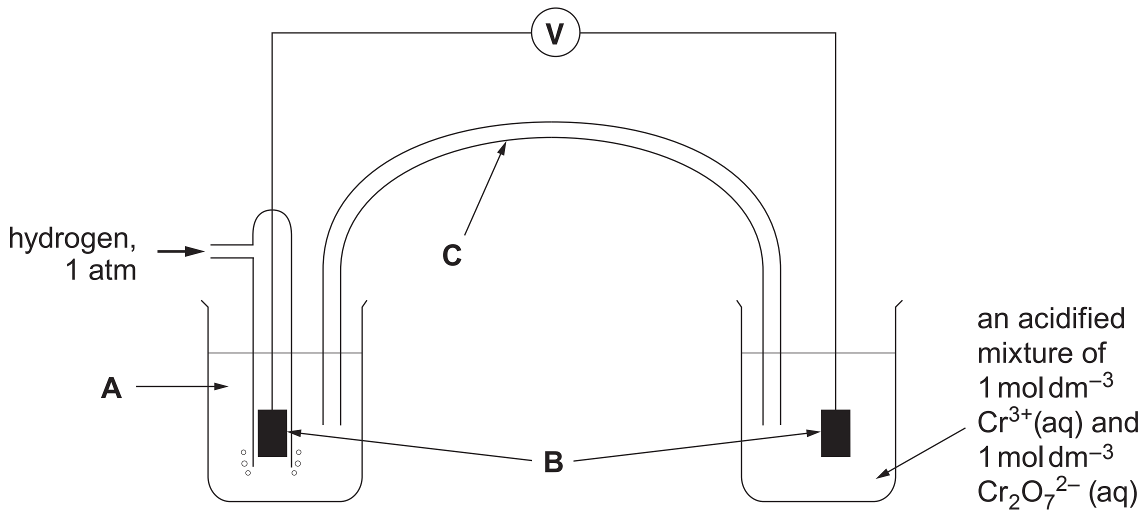
....

Examiner only

(e) One way of assessing whether a reaction is feasible is to use standard electrode potentials.

	Standard electrode potential / V
$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$	+1.77
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33

(i) The apparatus below can be used to measure the standard electrode potential for the $Cr_2O_7^{2-}/Cr^{3+}$ half-cell.



I. State what is represented by **A** and **B** on the diagram. [1]

A ....

B ....

II. **On the diagram** show the direction of flow of electrons in the external circuit. [1]

Examiner only

III. State what is represented by **C** on the diagram and state its function. [1]

IV. The concentrations of both $Cr_2O_7^{2-}$ and Cr^{3+} ions are $1\,mol\,dm^{-3}$. State and explain how the value shown on the high resistance voltmeter would change if the concentration of the Cr^{3+} ions were increased whilst the concentration of the $Cr_2O_7^{2-}$ was left unchanged. [2]

(ii) It is suggested that hydrogen peroxide could be used to oxidise Cr^{3+} ions in acidic solution to form dichromate ions.

I. Write an equation for this proposed reaction. [1]

II. Use the standard electrode potential values given to predict whether this reaction is feasible. [2]

III. Another method of finding whether a reaction is feasible is to use the Gibbs free energy calculated from standard enthalpy of formation and standard entropy values.

State, giving a reason, whether Gibbs free energies or electrochemical methods are more appropriate for finding whether this reaction is feasible. [2]

18

12. A student was provided with five ionic solids containing familiar cations and anions. Each solid contains a different **cation**. The solids were labelled **A**, **B**, **C**, **D** and **E**.

The student attempted to dissolve all five solids in water. She then attempted to dissolve those that did not dissolve in water in a stoichiometric amount of acid.

The results are given below.

Solid	Addition of water	Addition of dilute sulfuric acid to solid	Addition of dilute nitric acid to solid
A	dissolves giving colourless solution **1**		
B	dissolves giving colourless solution **2**		
C	does not dissolve	pale blue solution **3** forms upon warming with the acid	
D	does not dissolve	effervescence and solution **4** forms	
E	does not dissolve	some effervescence but the solid does not dissolve	effervescence and solution **5** forms

Pairs of the solutions formed were mixed and the observations recorded.

	Solution **5**	Solution **4**	Solution **3**	Solution **2**
Solution **1** (formed by dissolving solid **A**)	no visible change	white precipitate	white precipitate	bright yellow precipitate
Solution **2** (formed by dissolving solid **B**)	no visible change	no visible change	brown solution with a white solid	
Solution **3** (formed by reacting solid **C** with acid)	thick white precipitate	no visible change		
Solution **4** (formed by reacting solid **D** with acid)	thick white precipitate			

Flame tests

Flame tests carried out on solutions **1**, **2**, **3**, **4** and **5** gave no colour with one solution, an apple-green flame with another and a lilac flame with a third. The student noted unfamiliar flame test colours for the other two solutions.

Examiner only

(a) Suggest which anion is present in substances **D** and **E**. Give a reason for your answer. [2]

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(b) Use **all** the information provided to suggest identities for compounds **A**-**E**.

Explain clearly how you identified compounds **A** and **C**. [6 QER]

A ....

B ....

C ....

D ....

E ....

Explanation

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8

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL

1410U30-1

S19-1410U30-1

TUESDAY, 4 JUNE 2019 – AFTERNOON

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 8.**	**10**	
Section B	**9.**	**15**	
	10.	**15**	
	11.	**12**	
	12.	**15**	
	13.	**13**	
	Total	**80**	

1410U301 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11***(b)*.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

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Examiner only

SECTION A

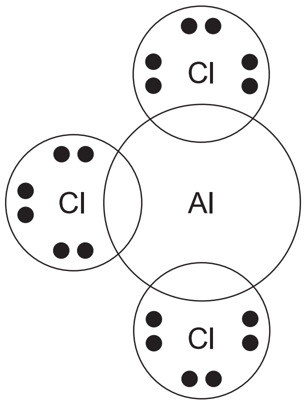
*Answer **all** questions in the spaces provided.*

1. Complete the electronic structure of the Co^{2+} ion below. [1]

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ ....

2. The compound $AlCl_3.NH_3$ contains both covalent and coordinate bonds.

Complete the dot and cross diagram for this compound. [2]



3. Write the expression for the ionic product of water, K_w. [1]

4. The standard hydrogen electrode and hydrogen fuel cells both use the same metal as an electrode. Name the metal and explain why this metal is suitable. [1]

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Examiner only

5. When CCl_4 is added to water two separate layers are formed and there is no reaction, whilst $SiCl_4$ reacts violently with water.

Give the reason for this difference. [1]

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6. The entropy change of the reaction shown below is positive. Give a reason for this. [1]

$$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

....

....

7. Carbon monoxide can be used to extract iron by the reduction of Fe_3O_4.

(a) Write an equation for this reaction. [1]

....

(b) Give a reason why carbon monoxide acts as a reducing agent while the equivalent oxide of lead does not. [1]

....

....

8. Draw the **structure** of the copper-containing ion formed when concentrated hydrochloric acid is added to a solution containing $[Cu(H_2O)_6]^{2+}$ ions. [1]

10

1410U301 03

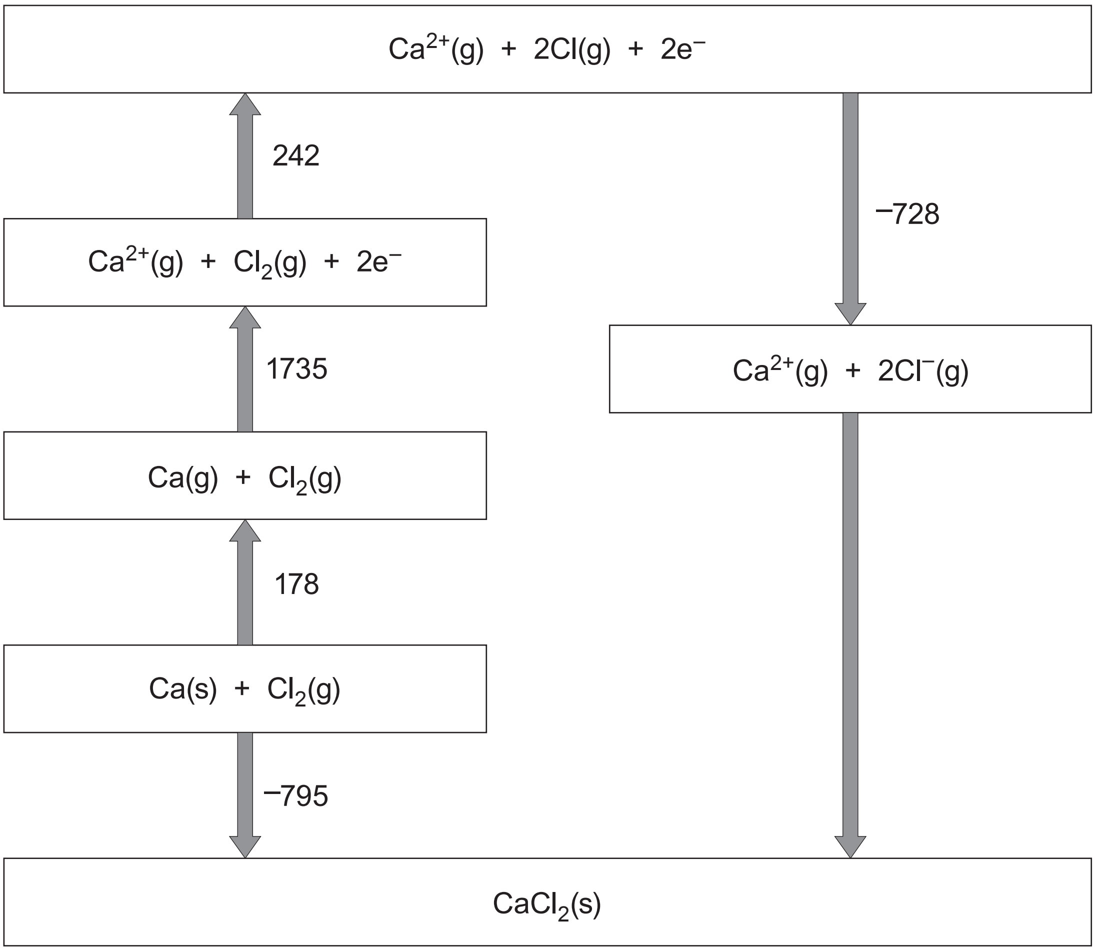
SECTION B

Answer **all** *questions in the spaces provided.*

9. Calcium chloride is an ionic solid that is used as a drying agent due to its ability to absorb water and form a range of hydrated crystals.

(a) The Born-Haber cycle below shows the formation of calcium chloride from its elements.

All values shown are standard enthalpy changes and are given in $kJ\,mol^{-1}$.



Examiner only

(i) Give the value of the standard enthalpy change of atomisation of chlorine. [1]

.... $kJ\,mol^{-1}$

(ii) Calculate the standard enthalpy of lattice formation of $CaCl_2$. [2]

$\Delta_{latt\ form}H^{\theta}$ = ... $kJ\,mol^{-1}$

(iii) A student incorrectly states that the second ionisation energy of calcium is 1735 $kJ\,mol^{-1}$ as this is the energy needed to form Ca^{2+}(g) in the Born-Haber cycle.

Suggest a possible value for the second ionisation energy of calcium, giving a reason for your answer. [2]

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Examiner only

(b) The reaction occurring when calcium chloride is used as a drying agent is shown below.

$$CaCl_2 + xH_2O \longrightarrow CaCl_2.xH_2O$$

(i) A laboratory manual suggests a **minimum** temperature of 200 °C to remove all the water from the hydrated calcium chloride. At this temperature the reaction occurring is as follows.

$$CaCl_2.xH_2O \longrightarrow CaCl_2 + xH_2O$$

A student calculates the enthalpy change when calcium chloride becomes dehydrated as 78.0 kJ mol^{-1}.

Assuming that this data is correct, calculate a value for the entropy change when calcium chloride becomes dehydrated at this temperature. [3]

ΔS = ... J K^{-1} mol^{-1}

(ii) In the drying process 5.20 g of hydrated calcium chloride forms 3.15 g of anhydrous calcium chloride.

Calculate the value of x in the original $CaCl_2.xH_2O$. [3]

x = ...

(iii) State the colour observed in a flame test using calcium chloride. [1]

....

Examiner only

(c) Calcium halides react with concentrated sulfuric acid in a similar manner to sodium halides.

State the observation(s) when concentrated sulfuric acid is added to calcium chloride and calcium bromide. Explain why the observations are different. [3]

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1410U301 07

Examiner only

10. Iodomethane, CH_3I, reacts with a range of nucleophiles. The rate of its reaction with thiosulfate ions, $S_2O_3^{2-}$, has been studied at a range of temperatures.

(a) The reaction is found to be first order with respect to iodomethane and has a rate constant of 3.10 $mol^{-1}\,dm^3\,s^{-1}$ at a particular temperature.

(i) Write the rate equation for this reaction. [2]

....

(ii) The activation energy for this reaction is 81.1 $kJ\,mol^{-1}$ with a frequency factor, *A*, of $5.62 \times 10^{12}\,mol^{-1}\,dm^3\,s^{-1}$. Find the temperature used for this reaction. [3]

T = ... K

Examiner only

(b) One method of performing this experiment involves sampling and quenching.

(i) Explain what is meant by the term 'quenching' and why it needs to be carried out. [2]

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(ii) One method of measuring the concentration of thiosulfate ions in solution is by titration using a solution containing iodine. Suggest an indicator for this titration. [1]

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(iii) A student suggests using gravimetric analysis to measure the amount of iodide ions produced in the reaction by addition of Pb^{2+}(aq) ions to form a precipitate. Give the colour of this precipitate. [1]

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09

(c) Iodomethane is an intermediate in the conversion of methanol to ethanoic acid.

(i) One catalyst used for this reaction is the iridium-containing complex ion $[Ir(CO)_2I_2]^-$. It is an example of a homogeneous catalyst.

I. State what is meant by a *homogeneous* catalyst. [1]

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II. Explain why transition metal complexes such as $[Ir(CO)_2I_2]^-$ can act as homogeneous catalysts. [2]

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Examiner only

(ii) Ethanoic acid is a weak acid with $K_a = 1.76 \times 10^{-5}$ mol dm^{-3}. Calculate the pH of an aqueous solution of ethanoic acid with a concentration of 0.220 mol dm^{-3}. [3]

pH = ...

15

Examiner only

11. Oxalic acid is the common name for ethanedioic acid. It is found in a wide range of plants and when purified it can form a dihydrate, $(COOH)_2.2H_2O$. The concentration of a saturated solution of oxalic acid can be found by titration.

(a) A sample of 25.0 cm^3 of a saturated solution of oxalic acid is diluted to form 250.0 cm^3 of an aqueous solution. Samples of 25.0 cm^3 of this diluted solution were heated and titrated using acidified potassium manganate(VII) to oxidise the oxalic acid.

(i) The half-equation for the oxidation of oxalic acid is shown below.

$$(COOH)_2 \longrightarrow 2CO_2 + 2H^+ + 2e^-$$

State the half-equation for the reduction of manganate(VII) ions in acid solution and use it to write the equation for the oxidation of oxalic acid by acidified manganate(VII) ions. [2]

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Examiner only

(ii) The titration of 25.0 cm^3 samples of the diluted oxalic acid solution against a potassium manganate(VII) solution of concentration 0.0505 mol dm^{-3} gave the following results.

	1	2	3	4
Volume of $KMnO_4$ solution / cm^3	31.65	31.30	31.60	31.75

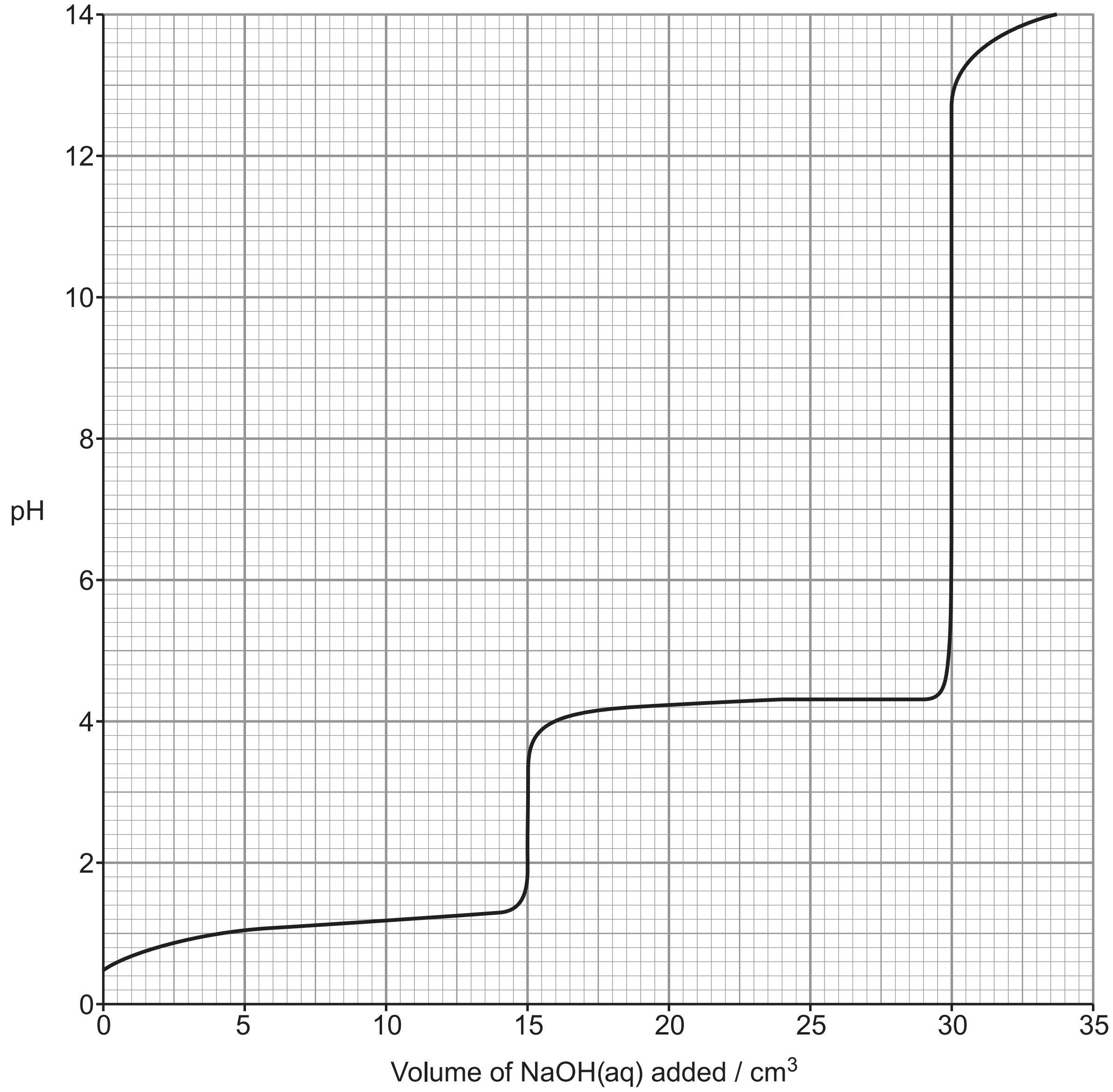
I. Use appropriate data to find the mean volume of $KMnO_4$ solution used. [1]

Mean volume = ... cm^3

II. Calculate the concentration of the original saturated solution of oxalic acid. [3]

Concentration = ... mol dm^{-3}

(b) Oxalic acid solutions may be analysed using acid-base titrations. The simplified pH curve for the titration of a solution of oxalic acid against sodium hydroxide is given below.



Four students undertook the titration in different ways:

- Alice used a pH probe to measure the pH after addition of each portion of sodium hydroxide solution
- Brychan used the indicator phenolphthalein in his titration; phenolphthalein has a pH range of 8.2 to 10.0
- Carys used the indicator methyl orange in her titration; methyl orange has a pH range of 3.2 to 4.4
- David used the indicator cresol red in his titration; cresol red has a pH range of 1.8 to 2.8

Examiner only

State and explain which, if any, of the methods chosen will allow the students to find valid data to calculate the concentration of oxalic acid. Briefly explain why oxalic acid has a pH curve of this shape. [6 QER]

You are not required to carry out any calculations.

12

Examiner only

12. *(a)* A student planned to distinguish between the following eight compounds.

magnesium hydroxide	magnesium carbonate
iron(II) hydroxide	iron(II) carbonate
chromium(III) hydroxide	chromium(III) carbonate
lead(II) hydroxide	lead(II) carbonate

He used the following method.

Step 1 Add dilute acid until all the solid has disappeared. Record any effervescence.

Step 2 Add 1 cm^3 of sodium hydroxide solution to each solution formed in step 1. Record any precipitate observed.

(i) State which compound(s) would give effervescence and why they do so. [2]

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(ii) The student plans to use dilute hydrochloric acid in step 1. His teacher tells him that this is **not** the correct acid to use.

Explain why hydrochloric acid should not be used and suggest an appropriate acid to use in its place. [2]

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Examiner only

(iii) Give the colours of the precipitates formed when sodium hydroxide solution is added to solutions containing the following ions. [2]

$Mg^{2+}(aq)$ ....

$Fe^{2+}(aq)$ ....

$Cr^{3+}(aq)$ ....

$Pb^{2+}(aq)$ ....

(iv) The method given opposite is incomplete.

Suggest an additional step that would allow the remaining solutions formed in step 2 to be identified. Give the expected observations. State the property of the metals that allows them to be distinguished in this way. [3]

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Examiner only

(b) Many rocks that contain carbonate anions also contain a mixture of cations. Atomic absorption spectroscopy can be used to find the ratio of the amounts of different cations present.

Analysis of an acid solution formed from the carbonate mineral huntite shows that it contains 91 $ng\,cm^{-3}$ of magnesium ions and 50 $ng\,cm^{-3}$ of calcium ions. These are the only two metal cations present.

(i) Find the formula of the mineral huntite. [3]

Formula ....

(ii) The initial solution was prepared using 220 μg of huntite. Calculate the volume of aqueous acid used to form the solution for analysis. [3]

Volume = ... dm^3

15

Examiner only

13. *(a)* Potassium dichromate(VI) is used as a reagent in acid solution in a wide range of redox reactions. The dichromate(VI) ions can be reduced using **excess** metallic zinc in acid conditions.

- Elfed performs the experiment in a sealed tube under an atmosphere of pure nitrogen gas
- Fatima performs the experiment in a standard test tube

The final products are different in each case. Use the standard electrode potentials below to identify the products in both cases. Give reasons for your answers. [4]

	Standard electrode potential, E^{θ}/V
$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$	–0.78
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	–0.76
$Cr^{3+}(aq) + e^- \rightleftharpoons Cr^{2+}(aq)$	–0.42
$O_2(g) + 2H_2O(l) + 4e^- \rightleftharpoons 4OH^-(aq)$	+0.40
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l)$	+1.33

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Examiner only

(b) Chromium(VI) is present in the anions dichromate(VI), $Cr_2O_7^{2-}$, and chromate(VI), CrO_4^{2-}. These ions can be interconverted in solution with the following equilibrium being established.

$$Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$$

The equilibrium constant, K_c, for this reaction is $2.42 \times 10^{-15}\,mol^2\,dm^{-6}$.

In general, dichromate(VI) is the main species in acid solution and chromate(VI) is the main species in alkaline solution.

(i) Write an expression for the equilibrium constant, K_c, for this reaction. [1]

(ii) The mass of the water in $1.00\,cm^3$ of a dilute aqueous solution is 1.00 g. Show that the concentration of water in any dilute aqueous solution is $55.5\,mol\,dm^{-3}$. [2]

Examiner only

(iii) A solution was prepared containing equal concentrations of $Cr_2O_7^{2-}$ and CrO_4^{2-}. The concentration of each was 0.057 mol dm^{-3}. A student predicted that a solution containing equal amounts of both must be neutral. Is the student correct? Use calculation(s) to justify your answer. [4]

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(iv) The value of the equilibrium constant for this reaction increases when the mixture is warmed. State and explain what information this provides about the enthalpy change of the reaction. [2]

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13

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1410U30-1

Z22-1410U30-1

THURSDAY, 9 JUNE 2022 – AFTERNOON

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

1 hour 45 minutes

	For Examiner's use only		
	Question	Maximum Mark	Mark Awarded
Section A	1. to 7.	10	
Section B	8.	14	
	9.	20	
	10.	12	
	11.	24	
	Total	80	

1410U301 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**(c).

Examiner only

SECTION A

Answer **all** questions.

1. Addition of aqueous lead(II) nitrate to aqueous potassium iodide causes a precipitate to form.

(a) Give the colour of the precipitate. [1]

....

(b) Write an ionic equation for the reaction. [1]

....

2. Addition of excess hydrochloric acid to aqueous copper(II) sulfate causes the solution to turn green.

Give the formula of the copper-containing species present. [1]

....

3. The rate equation for the decomposition of N_2O_5 to form O_2 and NO_2 is first order overall.

(a) Write the rate equation for this reaction. [1]

....

(b) Suggest a rate-determining step for this reaction. [1]

....

4. State what is meant by the term ‘standard electrode potential’. Include the conditions required. [2]

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Examiner only

5. Phosphorus is able to form two different chlorides, PCl_3 and PCl_5. Nitrogen is only able to form one chloride, NCl_3.

Explain this difference. [1]

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6. Write the expression for the ionic product of water, K_w. [1]

7. Give a reason why the entropy of Hg(l) is greater than that of Au(s). [1]

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03

10

Examiner only

SECTION B

Answer **all** questions.

8. The Mohs scale measures the hardness of different materials and runs from 1 to 10, with 10 being the hardest. A fingernail is rated as 2.5 on the Mohs scale so any material that can be scratched with a fingernail has a hardness of less than 2.5. Graphite and the minerals halite and tachyhydrite, can all be scratched with a fingernail.

(a) The hardness value of graphite is approximately 1.5.

Describe the structure of graphite and explain why it is soft. [2]

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05

Examiner only

(b) Halite is the mineral name for rock salt. It contains sodium chloride.

The equation corresponding to the standard enthalpy change of formation of sodium chloride is as follows.

$$Na(s) \quad + \quad ½Cl_2(g) \longrightarrow NaCl(s) \qquad \Delta_f H^\theta = -411\ kJ\ mol^{-1}$$

Some standard enthalpy changes are given in the table.

Enthalpy term	Standard molar enthalpy change / $kJ\ mol^{-1}$
first ionisation energy of sodium	494
electron affinity of chlorine	–364
bond energy of Cl_2	242
enthalpy of atomisation of sodium	109

(i) State the enthalpy of atomisation of chlorine. [1]

....

Examiner only

(ii) Use the standard enthalpy changes to find the standard enthalpy change of lattice formation of sodium chloride. [3]

Standard enthalpy change of lattice formation = ... $kJ\,mol^{-1}$

(iii) A student suggests that the entropy change for the formation of sodium chloride must be positive as the reaction occurs easily.

Is the student correct? Justify your answer. [2]

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Examiner only

(c) Tachyhydrite is a soft, soluble mineral that has a general formula of $A_aB_bX_c.dH_2O$. A and B represent two different s-block metal ions, with X representing an anion.

Use **all** the following data to find the identity of all the ions present and to find the formula of tachyhydrite.

A, B and X are not the chemical symbols of the elements present.

A flame test on a sample of tachyhydrite gives a brick-red flame with no indication of any other colour. Addition of dilute sulfuric acid to a solution of tachyhydrite causes the solution to go cloudy. With concentrated sulfuric acid, misty fumes are released from the solid, however there is no colour change. Addition of aqueous silver nitrate to a solution of tachyhydrite causes a precipitate to form.

Tachyhydrite has an M_r of approximately 518. On heating to constant mass the solid loses 41.75 % of its initial mass. [6 QER]

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Formula of tachyhydrite ....

14

Examiner only

9. Iron is an example of a transition element. It has two main oxidation states in its compounds.

(a) Write the electronic structure of an Fe^{2+} ion and use it to show why iron is classed as a transition element. [2]

....

....

(b) Explain why iron has more than one common oxidation state in its compounds. [1]

....

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(c) In very acidic solutions, Fe^{3+} forms $[Fe(H_2O)_6]^{3+}$.

Draw the structure of the $[Fe(H_2O)_6]^{3+}$ ion. [1]

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Examiner only

(d) In solutions with pH values between 1 and 3, the following equilibrium occurs.

$$[Fe(H_2O)_6]^{3+} \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H^+$$

The concentrations of $[Fe(H_2O)_6]^{3+}$ and $[Fe(H_2O)_5(OH)]^{2+}$ present can be studied using colorimetry.

(i) Explain why the complex ions $[Fe(H_2O)_6]^{3+}$ and $[Fe(H_2O)_5(OH)]^{2+}$ are not the same colour. [2]

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(ii) Suggest how you would select an appropriate wavelength to find the concentration of $[Fe(H_2O)_6]^{3+}$ in the equilibrium mixture. [1]

....

....

(iii) Write an expression for the equilibrium constant K_c for this equilibrium, giving its unit. [2]

Unit ....

Examiner only

(iv) A known mass of iron(III) chloride, $FeCl_3.6H_2O$ (M_r 270.4) is dissolved in 1.00 dm^3 of dilute hydrochloric acid. At equilibrium the pH of the solution is 1.55 and the concentration of $[Fe(H_2O)_5(OH)]^{2+}$ is 0.103 mol dm^{-3}. The numerical value of K_c under these conditions is 4.03×10^{-3}.

Find the mass of $FeCl_3.6H_2O$ used to make the solution. [4]

Mass = ... g

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Examiner only

(e) At higher pH values, solutions containing hydrated Fe^{3+} ions form a precipitate.

(i) Give the formula of the precipitate and its colour. [1]

Formula ....

Colour ....

(ii) Aqueous iron(II) compounds form a precipitate of a different colour with aqueous sodium hydroxide. In a test tube, the precipitate can change colour on standing. Use the standard electrode potential values below to explain this change. [2]

	Standard electrode potential, E^{θ} / V
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77
$O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$	+1.23

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(iii) A student states that it would be incorrect to use the standard electrode potential for the oxygen half-cell as the solution will be alkaline. State, giving a reason, the effect an alkaline solution will have on the value of the electrode potential of the oxygen half-cell. [2]

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Examiner only

(f) In a blast furnace Fe_2O_3 is reduced by CO to form iron metal.

(i) Write an equation for this process. [1]

....

(ii) Explain why CO is a good reducing agent. [1]

....

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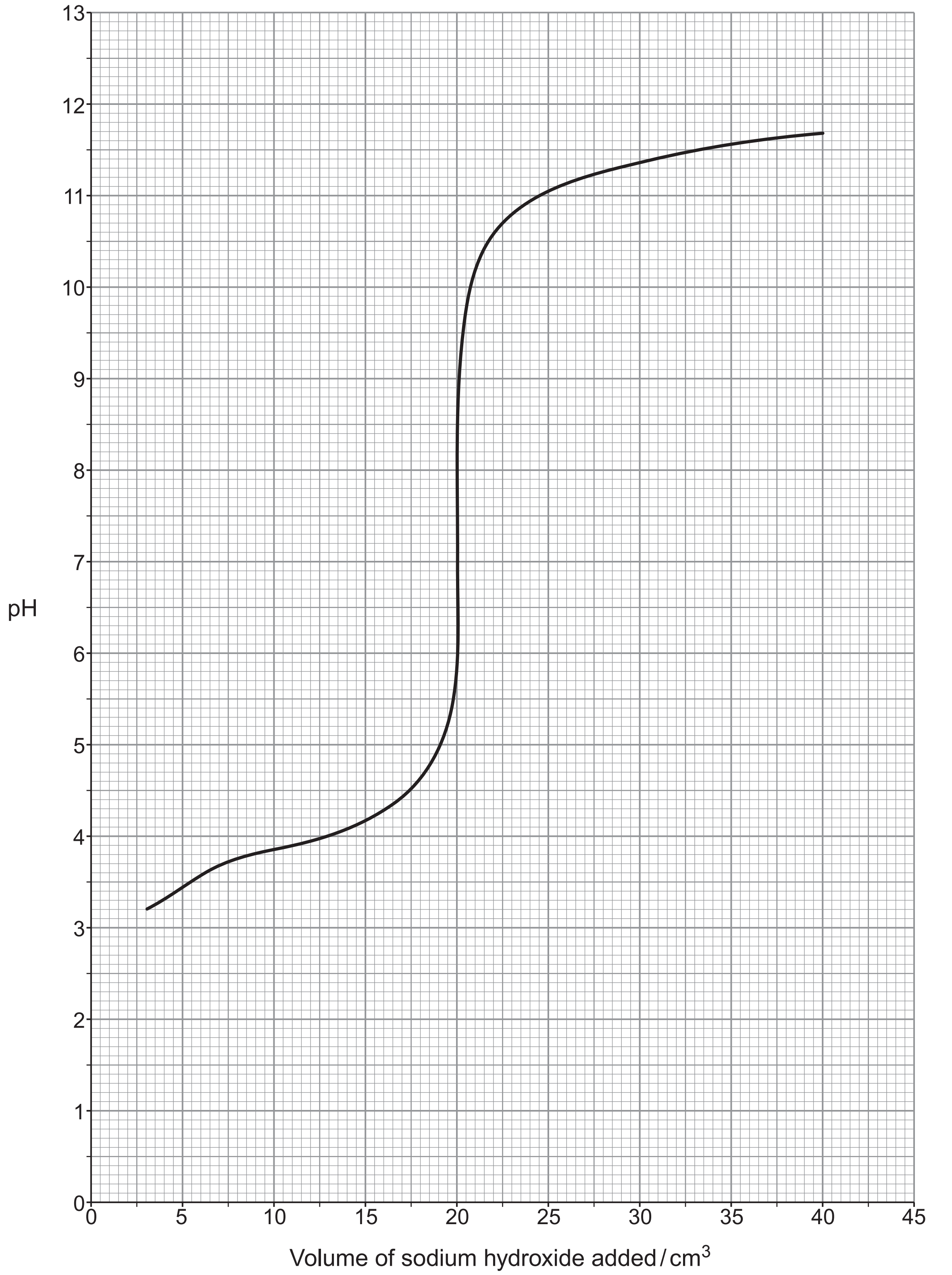
20

1410U301
13

10. A student used a solution of weak acid HA in a titration. A 25.0 cm^3 sample of the acid was titrated against a sodium hydroxide solution of concentration 0.250 mol dm^{-3} giving the pH titration curve shown. The initial pH value is missing from the graph.



Examiner only

(a) Several indicators are listed in the table.

Identify **all** the indicator(s) which are appropriate for this titration. Give a reason for your choice(s). [2]

Indicator	pH range
methyl orange	3.1 - 4.4
bromocresol purple	5.2 - 6.8
bromothymol blue	6.0 - 7.6
naphtholphthalein	7.3 - 8.7
cresolphthalein	8.2 - 10.1

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(b) Calculate the concentration of the HA solution. [2]

Concentration = ... $mol\,dm^{-3}$

Examiner only

(c) Find the pH of the mixture after addition of $10.0\,cm^3$ of sodium hydroxide solution and hence calculate the initial pH of the HA solution. [4]

You **must** show your working.

pH after addition of $10.0\,cm^3$ = ...

Initial pH = ...

(d) HA is a weak acid and so it can be used to make a buffer solution.

Suggest **one** important use for buffer solutions. [1]

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Examiner only

(e) Some data regarding the dissociation of another weak acid, HX, at two different temperatures are given in the table.

$$HX \rightleftharpoons H^+ + X^-$$

	HX at 298 K	HX at temperature *T*
Percentage dissociated	3.9	3.4
ΔG for acid dissociation / $kJ\,mol^{-1}$	15.9	16.7
ΔH for acid dissociation / $kJ\,mol^{-1}$	–10.0	
ΔS for acid dissociation / $J\,K^{-1}\,mol^{-1}$	–87.0	

Find temperature *T* and hence explain in terms of equilibrium why the percentage dissociation is different at this temperature. [3]

T = ... K

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Examiner only

11. Chlorine can form a range of oxyanions with chlorine present in oxidation states including +1, +5 and +7.

(a) Chlorate(I) ions, ClO^-, are commonly present in household bleaches. These ions can oxidise iodide ions to iodine.

$$ClO^- + 2I^- + H_2O \longrightarrow I_2 + Cl^- + 2OH^-$$

Dewi took a 25.0 cm^3 sample of a household bleach and diluted it to form 250 cm^3 of solution in a standard volumetric flask. Four 25.0 cm^3 portions of the diluted solution were measured and placed in separate conical flasks and a spatula of solid potassium iodide was added to each. Each portion was titrated against standard sodium thiosulfate solution from a burette.

$$2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$$

Dewi had access to five concentrations of sodium thiosulfate: 2.00 mol dm^{-3}, 1.00 mol dm^{-3}, 0.500 mol dm^{-3}, 0.200 mol dm^{-3} and 0.0500 mol dm^{-3}.

He selected the 0.500 mol dm^{-3} solution and his results are shown in the table.

	1	2	3	4
Volume of $Na_2S_2O_3$ solution / cm^3	6.45	6.40	6.50	6.45

(i) Calculate the mean volume of sodium thiosulfate solution used. [1]

Mean volume = ... cm^3

Examiner only

(ii) Find the concentration of sodium chlorate(I) in the initial bleach sample. [3]

Concentration = ... $mol\,dm^{-3}$

(iii) Bleach concentrations are often quoted on labels as % w/v, that is the mass of sodium chlorate(I) dissolved in $100\,cm^3$ of water.

Calculate the % w/v concentration of sodium chlorate(I) in this bleach. [1]

Concentration = ... % w/v

(iv) The teacher tells Dewi that he has selected an inappropriate concentration of sodium thiosulfate solution.

Suggest which concentration of sodium thiosulfate he should have chosen. Give **two** reasons for your answer. [2]

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Examiner only

(b) Chlorate(V) anions, ClO_3^-, are strong oxidising agents and can oxidise bromide ions in acid solution to form bromine and chloride ions.

$$ClO_3^- \;+\; 6Br^- \;+\; 6H^+ \longrightarrow Cl^- \;+\; 3Br_2 \;+\; 3H_2O$$

The initial rate of this reaction was measured at a temperature of 298 K using different concentrations of reactants.

Experiment	Concentration of ClO_3^- / mol dm^{-3}	Concentration of Br^- / mol dm^{-3}	pH	Initial rate / mol dm^{-3} s^{-1}
1	0.30	0.20	0	3.06×10^{-7}
2	0.60	0.20	0	6.12×10^{-7}
3	0.30	0.60	0	9.18×10^{-7}
4	0.30	0.60	1	

(i) I. Find the orders of reaction with respect to chlorate(V) and bromide ions. [2]

You **must** show your working.

Order with respect to chlorate(V) ions ....

Order with respect to bromide ions ....

Examiner only

II. The student finds that the rate of reaction is proportional to $[H^+]^3$. This is described as third order with respect to hydrogen ions.

Find the expected initial rate for experiment 4. [2]

Initial rate = ... $mol\,dm^{-3}\,s^{-1}$

(ii) Chemists often use an approximate rule that the rate of a reaction doubles when the temperature is increased by 10 °C.

Show that this rule is true if this reaction is warmed from 298 K to 308 K. The activation energy for the reaction is 52.8 kJ mol^{-1}. [3]

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Examiner only

(c) The acid formed from chlorate(VII) ions is commonly called perchloric acid, $HClO_4$. It is a very strong acid and is commonly classed as a superacid as it is a stronger acid than sulfuric acid.

(i) State how the K_a value of a stronger acid will compare with that of a weaker acid, giving a reason. [1]

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(ii) The concentration of 25.0 cm^3 of an aqueous solution of perchloric acid can be found by adding an excess of calcium carbonate and then measuring the amount of carbon dioxide gas released.

This can be done by either:

- measuring the volume of the gas released using a gas syringe, or
- measuring the mass lost as the carbon dioxide is released by placing the reaction flask on a digital balance.

Joe uses the first method to study his sample of perchloric acid and finds that the reaction releases 87 cm^3 of carbon dioxide at 30 °C and 1 atm pressure.

Heledd uses the second method and finds that the mixture loses a total of 0.1533 g during the reaction.

Examiner only

I. Find the number of moles of carbon dioxide released in each experiment and hence state whether the two solutions provided to Joe and Heledd are of the same concentration. [4]

You **must** show your working and give your answers to **appropriate** numbers of significant figures.

Moles of CO_2 using Joe's method = ... mol

Moles of CO_2 using Heledd's method = ... mol

....

....

II. Suggest, from the data provided, which of the two methods will give better results. Give a reason for your suggestion. [1]

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....

III. Isha suggests using excess magnesium in place of calcium carbonate for Joe and Heledd's experiments. This will produce hydrogen gas in place of carbon dioxide. Explain which of the two methods will suffer the greater loss in accuracy. [2]

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Examiner only

(iii) One common salt of perchloric acid is ammonium perchlorate.

Suggest a pH value for a solution of ammonium perchlorate. Give a reason for your answer. [2]

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1410U30-1

S23-1410U30-1

MONDAY, 12 JUNE 2023 – MORNING

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**14**	
	9.	**13**	
	10.	**18**	
	11.	**11**	
	12.	**14**	
	Total	**80**	

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01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q9**(a).

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Examiner only

SECTION A

Answer **all** questions.

1. Addition of aqueous lead nitrate to aqueous potassium iodide causes a precipitate to form. Write an ionic equation for the formation of the precipitate, including state symbols. [1]

....

2. Use the data shown to explain why silver chloride is insoluble in water. [2]

	Enthalpy / $kJ\,mol^{-1}$
Standard enthalpy of hydration for Ag^+	−464
Standard enthalpy of hydration for Cl^-	−364
Enthalpy of lattice formation of AgCl	−905

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3. A white crystalline solid gives a red colour in a flame test and addition of concentrated sulfuric acid to the solid gives purple fumes and a smell of rotten eggs.
Name the ions present, giving your reasons. [2]

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4. Aluminium chloride forms Al_2Cl_6 dimers using coordinate bonding. Explain why the coordinate bonds form. [2]

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Examiner only

5. Addition of ammonia solution to aqueous copper(II) sulfate causes a pale blue precipitate to form followed by a coloured solution when excess ammonia solution is added.

Give the colour of the solution formed. [1]

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6. The Haber process uses a heterogeneous catalyst.
State what is meant by the term 'heterogeneous' in this context. [1]

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7. Hexagonal boron nitride is sometimes called white graphite. Give **one** difference between the physical properties of hexagonal boron nitride and graphite. [1]

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SECTION B

8. When performing redox titrations the concentration of one solution is found by reaction with another solution of known concentration. Some oxidising agents are classed as primary standards as their concentrations can be calculated precisely from the mass used. Other oxidising agents need to be standardised by reaction with solutions of known concentration before use.

(a) Potassium dichromate, $K_2Cr_2O_7$, is a primary standard. Calculate the concentration of a solution of potassium dichromate made from 5.00 g of potassium dichromate in 250 cm^3 of deionised water. [2]

concentration = ... $mol\,dm^{-3}$

(b) Sodium thiosulfate, $Na_2S_2O_3$, can be standardised using potassium dichromate. To do this, acidified potassium dichromate is used to oxidise iodide ions to iodine molecules and the sodium thiosulfate is then used to reduce the iodine formed.

$$I_2 + 2e^- \rightleftharpoons 2I^-$$

$$2S_2O_3^{2-} \rightleftharpoons S_4O_6^{2-} + 2e^-$$

(i) Write the half-equation for acidified dichromate ions acting as an oxidising agent. [1]

....

(ii) Write an equation for the oxidation of iodide ions to iodine molecules using acidified dichromate. [1]

....

Examiner only

(iii) Excess KI(aq) was added to $25.0\,cm^3$ of the aqueous potassium dichromate from part (a) with a small amount of dilute acid. This produced iodine. Aqueous sodium thiosulfate was added from a burette and $24.30\,cm^3$ of $Na_2S_2O_3(aq)$ was needed for complete reaction with the iodine.

I. The titration used an indicator. Name the indicator used. [1]

....

II. Calculate the concentration of the aqueous sodium thiosulfate. [3]

concentration = ... $mol\,dm^{-3}$

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Examiner only

(c) Addition of aqueous sodium hydroxide to the solution of potassium dichromate causes a colour change.
State the colour change and explain why this is not a redox reaction. [2]

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(d) Heating solid potassium dichromate can cause it to decompose releasing oxygen gas.

$$4K_2Cr_2O_7 \longrightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$$

$$\Delta H^\theta = 348\ \text{kJ mol}^{-1} \quad \Delta S^\theta = 385\ \text{J K}^{-1}\ \text{mol}^{-1}$$

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta$/kJ mol^{-1}
K_2CrO_4	−1382
Cr_2O_3	−1128

(i) Calculate the standard enthalpy change of formation of potassium dichromate, $K_2Cr_2O_7$. [2]

$\Delta_f H^\theta$ = ... kJ mol^{-1}

(ii) Calculate the minimum temperature required for this reaction. [2]

minimum temperature = ... K

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Examiner only

9. (a) The elements of the p-block show a variety of oxidation states within the same group, whilst some elements of the d-block can show an even wider range of oxidation states.

- Some oxides of group 4 have an oxidation state of +4 such as CO_2, SiO_2 and PbO_2 and others have an oxidation state of +2 such as CO and PbO. There is no stable oxide with the formula SiO.
- Some chlorides of group 5 have a +5 oxidation state such as PCl_5 and some have an oxidation state of +3 such as NCl_3 and PCl_3. There is no stable chloride with the formula NCl_5.
- Manganese forms a range of oxides including MnO, Mn_2O_3 and MnO_2 which have oxidation states of +2, +3 and +4 respectively.

Explain why these elements can form the oxidation states shown and why SiO and NCl_5 do not form. [6 QER]

Examiner only

(b) Tin(II) ions, Sn^{2+} (aq), can be used as reducing agents for a range of metal ions. In one study of the reduction of Pt^{4+} ions to Pt^{2+} ions in an acid solution, the reaction was found to be first order with respect to both Pt^{4+} and Sn^{2+} with a rate constant of 895 $mol^{-1} dm^3 s^{-1}$ at 297 K.

(i) Write a rate equation for this reaction. [1]

....

(ii) A student suggests studying this reaction using sampling and quenching with samples taken every 30 seconds for 5 minutes. Explain why this would not be an appropriate sampling interval. [2]

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(c) (i) A second student decides to use colorimetry to study the rate of this reaction. Explain what the student should consider when choosing an appropriate wavelength of light for their colorimetry experiment. [1]

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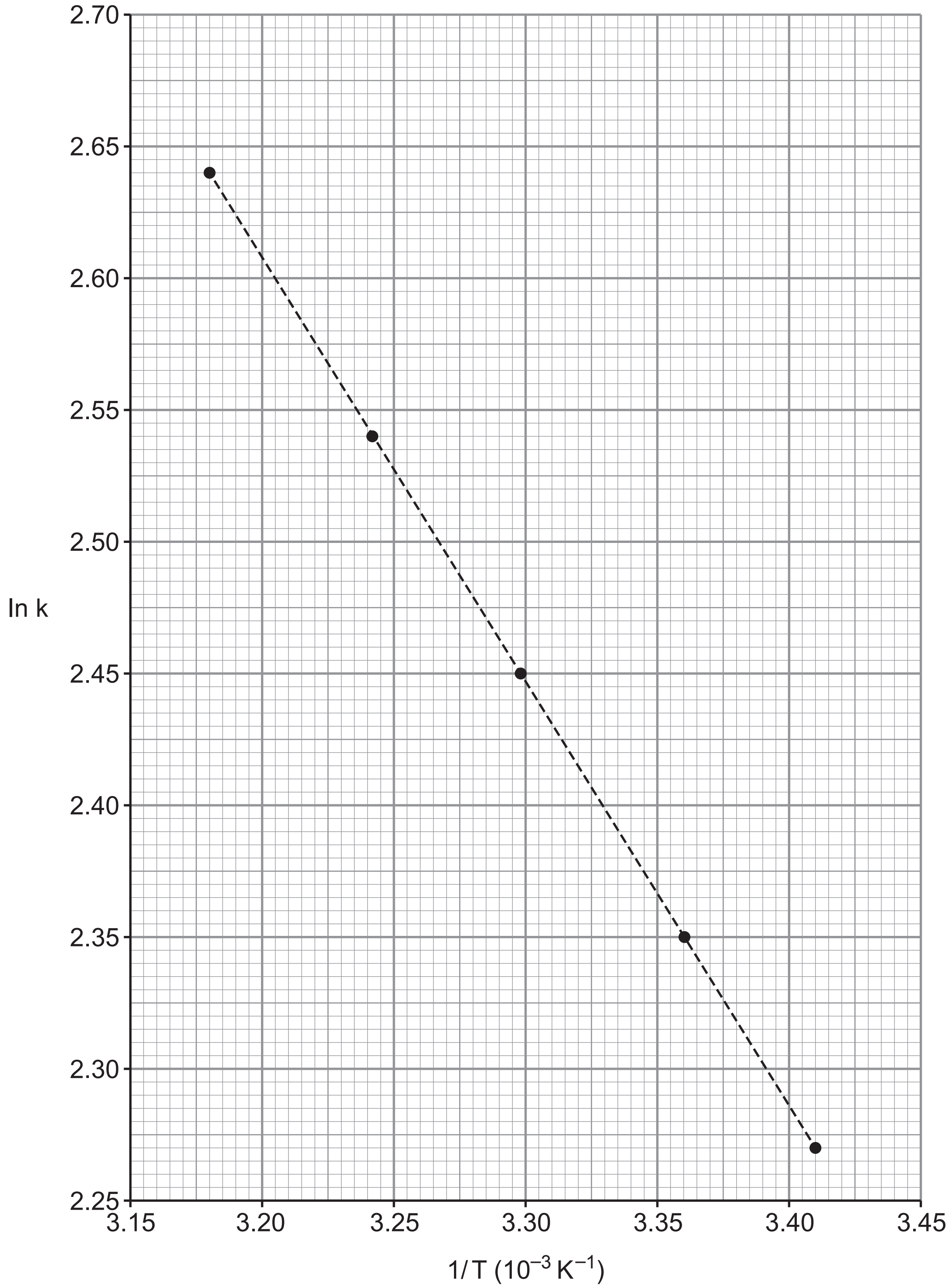
(ii) The value of the rate constant was measured at five different temperatures to attempt to find the activation energy.

$$\ln k = \ln A - (E_a/RT)$$

A graph was plotted of (ln k) against (1/Temperature) and this is shown below.



Examiner only

Find the activation energy of the reaction. [3]

activation energy, E_a = ... $kJ\,mol^{-1}$

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Examiner only

10. Oxygen forms several molecular compounds of formula X_2O. These include compounds with fluorine and chlorine.

(a) Some thermodynamic data for F_2O and Cl_2O are given below.

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / kJ\,mol^{-1}$	Standard molar entropy, $S^\theta / J\,K^{-1}\,mol^{-1}$
F_2O	24.5	247
Cl_2O	80.3	266
O_2		205
F_2		203
Cl_2	0	

A student examines the data and states that there is sufficient information to find the minimum temperature for decomposition of F_2O to its elements but not for Cl_2O.

Is the student correct? Justify your answer. [3]

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(b) F_2O is a strong oxidising agent and can oxidise Xe to XeF_4 whilst releasing oxygen gas.

(i) Write an equation for this process. [1]

....

(ii) Show that the xenon is oxidised in the process. [1]

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Examiner only

(c) F_2O reacts slowly with water to produce HF, a weak acid.

$$F_2O \quad + \quad H_2O \longrightarrow 2HF \quad + \quad O_2$$

A small amount of F_2O is added to 400 cm^3 of water at 15 °C and it reacts completely. The reaction produces 82 cm^3 of oxygen gas at this temperature and 1 atm pressure.

(i) Calculate the concentration of the HF solution formed. [2]

concentration = ... mol dm^{-3}

(ii) Calculate the pH of the HF solution formed. [3]

$$K_a \text{ (HF)} = 6.6 \times 10^{-4} \text{ mol dm}^{-3}$$

pH = ...

Examiner only

(d) Cl_2O can react with water in a reversible reaction.

$$Cl_2O \quad + \quad H_2O \rightleftharpoons 2\ HOCl$$

This reaction can occur in non-aqueous solvents and in a particular solvent the value of the equilibrium constant, K_c, at 273 K is 5.0 and it has no units.

(i) Write an expression for the equilibrium constant K_c. [1]

(ii) A student studies the equilibrium at a higher temperature. She adds samples of 0.40 mol of H_2O and 0.14 mol of Cl_2O to the non-aqueous solvent giving a total volume of 1000 cm^3.

At equilibrium the concentration of HOCl is 0.22 mol dm^{-3}.

I. Find the value of the equilibrium constant K_c at this temperature. [3]

K_c = ...

II. State, giving a reason, whether this reaction is endothermic or exothermic. [2]

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Examiner only

III. HOCl can act as an oxidising agent.

$$2HOCl\,(aq) + 2H^{+} + 2e^{-} \rightleftharpoons Cl_2\,(g) + 2H_2O\,(l) \qquad E^{\theta} = +1.63\ V$$

Give the formulae of all the metal ions in the table that can be oxidised by HOCl (aq). Explain why you have chosen these ions. [2]

	Standard electrode potential, E_{θ}/V
$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$	+0.77
$Tl^{3+} + 2e^{-} \rightleftharpoons Tl^{+}$	+1.25
$Pb^{4+} + 2e^{-} \rightleftharpoons Pb^{2+}$	+1.69
$Co^{3+} + e^{-} \rightleftharpoons Co^{2+}$	+1.82

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Examiner only

11. Carboxylic acids are classed as weak acids.

(a) State what is meant by the term weak acid. [1]

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(b) Substituted propanoic acids, $CH_3CHXCOOH$, have different acid strengths. The table below gives some information about three substituted propanoic acids.

Substituted propanoic acid, $CH_3CHXCOOH$	
X = H	The pH of a 0.50 mol dm^{-3} aqueous solution is 2.59
X = Cl	$K_a = 1.48 \times 10^{-3}$ mol dm^{-3}
X = OH	pK_a = 3.86

Place these acids in order of increasing acid strength. **You must show your working**. [3]

Weakest .... Strongest

Examiner only

(c) A buffer is produced by mixing 100 cm^3 of ethanoic acid ($K_a = 1.8 \times 10^{-5}$ mol dm^{-3}) of concentration 0.80 mol dm^{-3} with 50 cm^3 of aqueous sodium hydroxide of concentration 0.80 mol dm^{-3} at a temperature of 298 K.

(i) Calculate the initial pH of the aqueous sodium hydroxide. [2]

pH = ...

(ii) Calculate the pH of the buffer solution. [2]

pH = ...

(iii) Explain how buffers such as this work. [3]

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11

Examiner only

12. The formula of an unknown mineral is $Mg_aX_b(CO_3)_c(OH)_d{\cdot}eH_2O$.

A student performs a series of experiments to calculate the values **a**, **b**, **c**, **d** and **e** and name element X.

Experiment Number	Experiment	Result
1	Heating a sample of 0.0200 mol of the mineral to dryness.	The sample loses 1.44 g of mass.
2	Addition of 0.0200 mol of the mineral to excess hydrochloric acid.	A volume of 490 cm^3 of carbon dioxide gas is released at 298 K and 1 atm pressure. The remaining solution has a green colour.
3	Addition of excess aqueous sodium hydroxide gradually to the solution produced in experiment 2.	A precipitate forms that shows a mixture of white and grey-green colours. Addition of excess sodium hydroxide leaves a white precipitate in a green solution.
4	Addition of a sample of 1.00×10^{-3} mol of the mineral to 25.0 cm^3 of HCl of concentration 1.00 $mol\,dm^{-3}$. Titration of the remaining acid against standard aqueous sodium hydroxide of concentration 0.500 $mol\,dm^{-3}$.	A volume of 14.00 cm^3 of the aqueous sodium hydroxide is required for complete reaction.

(a) Find the value of **e**, the number of water molecules in each formula unit. [2]

e = ...

(b) Find the value of **c**, the number of carbonate ions in each formula unit. [2]

c = ...

Examiner only

(c) Explain the observations seen in experiment 3. Use these to name element X. [3]

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(d) In experiment 4 the acid reacts with all the hydroxide and carbonate ions.

$$H^+ + OH^- \longrightarrow H_2O$$

$$2H^+ + CO_3^{2-} \longrightarrow H_2O + CO_2$$

Calculate the value of **d**, the number of hydroxide ions present in each formula unit. [4]

d = ...

Examiner only

(e) The compound has 56 atoms in each formula unit.
Use your answers to (a), (b), (c) and (d), and the total number of atoms to find the formula of the mineral. [3]

Formula ....

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END OF PAPER